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|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| CONTACT<br>INFORMATION | Department of Mathematics<br>University of Wisconsin–Madison<br>480 Lincoln Drive<br>Madison, WI 53706, USA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>E-mail:</i> hhong78@wisc.edu<br><i>Web:</i> <a href="https://hyukpyohong.github.io">https://hyukpyohong.github.io</a> |
| APPOINTMENTS           | <b>Institute for Basic Science (IBS)</b> , Daejeon, South Korea<br>Visiting Research Fellow, Biomedical Mathematics Group May 2026–present<br><br><b>University of Wisconsin–Madison</b> , Madison, Wisconsin, USA<br>Van Vleck Assistant Professor, Department of Mathematics Aug. 2023–present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                          |
| EDUCATION              | <b>KAIST</b> , Daejeon, South Korea<br>Ph.D. in Mathematical Sciences Feb. 2018–Aug. 2023<br>Advisor: Jae Kyoung Kim<br>Thesis: Development of stochastic model reduction framework for analysis and inference of biochemical reaction networks<br>B.S. in Mathematical Sciences Mar. 2013–Feb. 2018                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                          |
| RESEARCH<br>INTERESTS  | <b>Fields:</b> Mathematical biology, ODEs, Stochastic processes, Bayesian statistics, Digital medicine<br><b>Topics:</b> Steady states of ODEs, stationary distributions of CTMCs, Koopman operator theory, MCMC methods, parameter estimation for non-Markovian stochastic models, metabolic control analysis, homeostasis and adaptation in biological systems, neurodegenerative disease-related alteration of human motor activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                          |
| PAPERS                 | †: (co-)1st author, *: (co-)corresponding author.<br>Note: Where no author is designated as the first author (†), names are listed in alphabetical order by last name, as is standard practice in mathematical journals.<br><br>In preparation or preprint: <ol style="list-style-type: none"> <li>18. Ella Buelling†, Wolfram Liebermeister*, <b>Hyukpyo Hong*</b>, Connections between structural sensitivity analysis and metabolic control analysis, <i>in preparation</i></li> <li>17. <b>Hyukpyo Hong</b>, Seewoo Lee, Necessary and sufficient conditions of the existence of a finite dimensional invariant subspace for polynomial ODEs, <i>in preparation</i></li> <li>16. Geonhee Ryu†, <b>Hyukpyo Hong*</b>, Daewook Kim*, Non-autonomous dynamics learning via Koopman-based Neural Networks, <i>in preparation</i></li> <li>15. Seewoo Lee†, Byung-Hak Hwang†, Hyojae Lim, Jihoon Hyun, Ilkyoo Choi, Yeachan Park, Jineon Baek, <b>Hyukpyo Hong</b>, Keewoo Lee, Jaeseong Heo, Hyungryul Baik, Chul-hee Lee*, Kyu-Hwan Lee, LEAN-GAP: A Dataset of Formalized Graduate Algebra Problems, <i>Submitted</i></li> <li>14. <b>Hyukpyo Hong</b>†, Qin Li, Matthew Colbrook, Hanbaek Lyu*, Finding Koopman Invariant Subspaces via Personalized PageRank, <i>Submitted</i>; [<a href="#">arXiv</a>]</li> <li>13. <b>Hyukpyo Hong</b>†*, Diego Rojas La Luz†, Gheorghe Craciun*, Ubiquitous asymptotic robustness in biochemical systems, <i>Under review</i>; [<a href="#">arXiv</a>]</li> </ol> Published or accepted: <ol style="list-style-type: none"> <li>12. Dongju Lim†, Kyeong Tae Ko†, <b>Hyukpyo Hong</b>, Hyojung Lee, Boseung Choi, Won Chang, Sunhwa Choi*, Jae Kyoung Kim*, A history-dependent approach for accurate initial condition estimation in epidemic models, <i>PLoS Computational Biology</i>, 2025; [<a href="#">journal</a>]</li> </ol> |                                                                                                                          |

11. **Hyukpyo Hong**<sup>†</sup>, Seokhwan Moon<sup>†</sup>, Yuji Hirono<sup>†\*</sup>, Jae Kyoung Kim\*, Topological criterion for robust perfect adaptation of reaction fluxes in biological networks, *iScience*, 2025; [[arXiv](#)]; [[journal](#)]
10. **Hyukpyo Hong**<sup>†</sup>, Eunjin Eom<sup>†</sup>, Hyojung Lee, Sunhwa Choi\*, Boseung Choi\*, Jae Kyoung Kim\*, Overcoming bias in estimating epidemiological parameters with realistic history-dependent disease spread dynamics, *Nature Communications*, 2024; [[journal](#)]
9. Hyeontae Jo<sup>†</sup>, **Hyukpyo Hong**<sup>†</sup>, Hyung Ju Hwang, Won Chang, Jae Kyoung Kim\*, Density physics-informed neural networks reveal sources of cell heterogeneity in signal transduction, *Patterns*, 2023; [[bioRxiv](#)]; [[journal](#)]
8. **Hyukpyo Hong**<sup>†</sup>, Mark Jayson Cortez, Yu-Yu Cheng, Hang J. Kim, Boseung Choi\*, Krešimir Josić\*, Jae Kyoung Kim\*, Inferring delays in partially observed gene regulation processes, *Bioinformatics*, 2023; [[bioRxiv](#)]; [[journal](#)]
7. **Hyukpyo Hong**, Bryan S. Hernandez, Jinsu Kim, Jae Kyoung Kim\*, Computational translation framework identifies biochemical reaction networks with special topologies and their long-term dynamics, *SIAM Journal on Applied Mathematics*, 2023; [[arXiv](#)]; [[journal](#)]
6. **Hyukpyo Hong**<sup>†</sup>, Ji Yun Noh<sup>†</sup>, Hyojung Lee, Sunhwa Choi, Boseung Choi, Jae Kyoung Kim\*, Eui-Cheol Shin\*, Modeling incorporating the severity-reducing long-term immunity: higher viral transmission paradoxically reduces severe COVID-19 during endemic transition, *Immune Network*, 2022; [[medRxiv](#)]; [[journal](#)]
5. Dae Wook Kim<sup>†</sup>, **Hyukpyo Hong**<sup>†</sup>, Jae Kyoung Kim\*, Systematic inference identifies a major source of heterogeneity in cell signaling dynamics: the rate-limiting step number, *Science Advances*, 2022; [[journal](#)]
4. Jaehyoung Hong<sup>†</sup>, Su Jung Choi<sup>†</sup>, Se Ho Park, **Hyukpyo Hong**, Victoria Booth, Eun Yeon Joo\*, Jae Kyoung Kim\*, Personalized sleep-wake patterns aligned with circadian rhythm relieve daytime sleepiness, *iScience*, 2021; [[bioRxiv](#)]; [[journal](#)]
3. Yun Min Song<sup>†</sup>, **Hyukpyo Hong**<sup>†</sup>, Jae Kyoung Kim\*, Universally valid reduction of multiscale stochastic biochemical systems with simple non-elementary propensities, *PLoS Computational Biology*, 2021; [[bioRxiv](#)]; [[journal](#)]
2. Mark Jayson Cortez<sup>†</sup>, **Hyukpyo Hong**, Boseung Choi\*, Jae Kyoung Kim\*, Krešimir Josić\*, Hierarchical Bayesian models for inference in biochemical reactions with delays, *Bioinformatics*, 2021; [[bioRxiv](#)]; [[journal](#)]
1. **Hyukpyo Hong**<sup>†</sup>, Jinsu Kim<sup>†</sup>, M. Ali Al-Radhawi, Eduardo D. Sontag, Jae Kyoung Kim\*, Derivation of stationary distributions of biochemical reaction networks via structure transformation, *Communications Biology*, 2021; [[bioRxiv](#)]; [[journal](#)]

#### BOOK CHAPTERS

1. **Hyukpyo Hong**, Boseung Choi, Jae Kyoung Kim\*, Beyond the Michaelis-Menten: Bayesian inference for enzyme kinetic analysis, *Computational Methods for Estimating the Kinetics Parameters of Biological Systems*, Methods in Molecular Biology; [[journal](#)]; [[link](#)]

#### RESEARCH GRANTS

2019–2023 National Research Foundation of Korea (NRF), NRF-2019-Fostering Core Leaders of the Future Basic Science Program/Global Ph.D. Fellowship Program, 2019H1A2A1075303, **PI** (120000)  
 Title: *Development of Stochastic Model Reduction Methodology and its Application to an Optogenetic System*

#### HONORS AND AWARDS

2025 Postdoctoral Excellence in Teaching Award, UW-Madison  
 2025 Excellent Dissertation Award, Korean Mathematical Society  
 2024 KSIAM Next Generation Researcher Award, Korean Society for Industrial and Applied Mathematics  
 2022 SIAM Student Travel Award, Society for Industrial and Applied Mathematics  
 2022 Blood Donor Hall of Fame (100 donations), Korean Red Cross

Global Ph.D. Fellowship (Full Tuition), National Research Foundation of Korea  
 2017 36th National Undergraduate Mathematics Competition Silver Award, Korean Mathematical Society  
 2016 Mirae Asset Global Exchange Scholarship, Mirae Asset Park Hyeon Joo Foundation  
 2014 33rd National Undergraduate Mathematics Competition Silver Award, Korean Mathematical Society  
 2014 Dean's List Award College of Natural Sciences, KAIST  
 2013 32nd National Undergraduate Mathematics Competition Silver Award, Korean Mathematical Society  
 National Scholarship for Science and Engineering (Full Tuition), Korea Student Aid Foundation

## TEACHING

### UW–Madison

- Spring 2026: [Instructor] MATH331: Introductory Probability
- Fall 2025: [Instructor] MATH331: Introductory Probability
- Spring 2025: [Instructor] MATH331: Introductory Probability
- Fall 2024: [Instructor] MATH340: Elementary Matrix and Linear Algebra
- Spring 2024: [Instructor] MATH320: Linear Algebra and Differential Equations
- Fall 2023: [Instructor] MATH320: Linear Algebra and Differential Equations

### KAIST

- Spring 2019: [Teaching Assistant] MAS201: Differential Equations and Applications
- Fall 2018: [Teaching Assistant] MAS201: Differential Equations and Applications, MAS480: Introduction to Mathematical Biology
- Spring 2018: [Teaching Assistant] MAS109: Introduction to Linear Algebra, MAS212: Linear Algebra

## MENTORING

### UW–Madison

- Jun. 2024 – present: Ella Buelling, Junior student, Edgewood High School  
Working on connections between the hypergraph representation of biochemical reaction networks and metabolic control analysis.

### KAIST

- Jun. 2023 – Mar. 2025: Seokhwan Moon, Undergraduate, Department of Mathematics, POSTECH  
Investigated structural conditions under which ODE steady-state solutions remain independent of system parameters. Published in *iScience*.
- Spring 2019: Minyoo Kim, Undergraduate, Department of Mathematics, KAIST  
Investigated the total quasi-steady-state approximation for a competitive system.  
Won a poster prize at the 2019 KSIAM Spring Conference and a prize in the URP final evaluation.

## INVITED TALKS

2026-04-28, Online, **Formal reaction kinetics and related questions**  
*Ubiquitous Asymptotic Robustness in Biochemical Systems* [\[video\]](#)  
 2026-02-26, Online, **Mathematics of Reaction Networks (MoRN) Seminar**  
*Ubiquitous Asymptotic Robustness in Biochemical Systems* [\[video\]](#)  
 2025-11-21, Durham, NC, USA, **Duke University Math Bio Seminar**  
*Identifying major sources of heterogeneity in antibiotic cell responses via Bayesian inference and Physics-informed neural networks*

2025-11-20, Durham, NC, USA, **NC State Univ. TPB Cluster**  
*Bridging Mathematical Theory and Biological Challenges: From intracellular networks to human-scale dynamics*

2025-11-19, Durham, NC, USA, **NC State Univ. Dept. of Mathematics**  
*Leveraging Koopman Operator Theory and Graph Representation for Deeper Insight into Complex Dynamical Systems*

2025-11-13, Seoul, Korea, **Seminar at Seoul National University**  
*Finite-Dimensional Representations of the Koopman Operator: Equation- and Data-Driven Approaches*

2025-10-31, Madison, WI, USA, **UW–Madison Applied and Computational Mathematics Seminar**  
*Finite-Dimensional Representations of the Koopman Operator: Equation- and Data-Driven Approaches*

2025-10-28, Online, **Midwest Mathematical Biology Seminar**  
*Ubiquitous Asymptotic Robustness in Biochemical Systems*

2025-10-01, Seoul, Korea, **Seminar at Korea University**  
*Leveraging Koopman Operator Theory and Graph Representation for Deeper Insight into Complex Dynamical Systems*

2025-09-27, Chicago, IL, USA, **2025 GLSIAM Meeting**  
*Digital biomarker of cognitive impairment: fractal patterns of human activity data*

2025-07-17, Edmonton, Canada, **SMB Annual Meeting**  
*Inferring delay distributions in partially observed biological processes*

2025-07-08, Madison, WI, USA, **SIAM Conference on Applied Algebraic Geometry**  
*Ubiquitous Asymptotic Robustness in Biochemical Systems*

2025-06-20, Busan, Korea, **Pusan National University Math Seminar**  
*Parameter inference in Markovian and non-Markovian models*

2025-06-04, Daejeon, Korea, **KAIST Brain & Cognitive Sciences Seminar**  
*Digital biomarker of cognitive impairment: fractal patterns of human activity data*

2025-06-02, Seoul, Korea, **Konkuk University Math Seminar**  
*Parameter inference in Markovian and non-Markovian models*

2025-05-17, Seoul, Korea, **KSIAM Spring Conference**  
*KSIAM Next Generation Researcher Award speech*

2025-05-15, Denver, CO, USA, **SIAM Conference on Applied Dynamical Systems**  
*Network Translation of a Stochastic Reaction Network to Reveal Stationary Distributions*

2025-03-03, Online, **Computational and Applied Math / Data Science Seminar, Arizona State University**  
*Koopman representation: Linear representation of nonlinear dynamics*

2024-10-25, Cincinnati, OH, USA, **Applied Math Seminar at the University of Cincinnati**  
*Koopman representation: Linear representation of nonlinear dynamics*

2024-10-07, Online, **Seminar at the University of the Philippines Diliman**  
*Accurate epidemiological parameters estimation with realistic history-dependent disease spread dynamics*

2024-08-01, Busan, Korea, **Seminar at Pusan National University**  
*Analysis of dynamical systems using “easier” representations*

2024-07-31, Daejeon, Korea, **Biomedical Mathematics Seminar at Institute for Basic Science**  
*Koopman representation: Linear representation—not an approximation—of nonlinear dynamics*

2024-07-04, Seoul, Korea, **SMB Annual Meeting**  
*Density-PINNs reveal that multiple pathways reduce cell-to-cell heterogeneity in antibiotic responses*

2024-07-02, Seoul, Korea, **SMB Annual Meeting**  
*Inferring delays in partially observed gene regulation processes*

2024-06-04, Sejong, Korea, **Data Computational Sciences Special Lecture at Korea University, Sejong Campus**  
*How to sample random numbers from a target distribution*

2024-06-03, Seoul, Korea, **Data Science Seminar at Sogang University**  
*Analysis of dynamical systems using “easier” representations*

2024-05-23, Pohang, Korea, **CM2LA Seminar at POSTECH**  
*Analysis of dynamical systems using “easier” representations*

2024-04-25, Champaign, IL, USA, **Mathematical Biology Seminar at UIUC**  
*Analyzing long-term behavior of dynamical systems using “easy-to-handle” representations*

2023-08-17, Daejeon, Korea, **ICIAM 2023 Satellite Workshop: Stochastic Modeling and Data Analysis for Biological Systems**  
*Network translation allows for revealing long-term dynamics of stochastic reaction networks*

2023-07-17, Columbus, OH, USA, **SMB Annual Meeting**  
*Network translation allows for revealing long-term dynamics of stochastic reaction networks*

2023-06-27, Jeju, Korea, **The 8th CIJK International Conference on Mathematical and Theoretical Biology**  
*Things we have been overlooking in infectious disease modeling*

2023-05-19, Pyeongchang, Korea, **KSIAM Spring Conference, Special session**  
*Increasing viral transmission paradoxically reduces severe COVID-19 during endemic transition*

2022-12-19, Daejeon, Korea, **Mini-workshop on Recent Trends in Pure and Applied Mathematics**  
*Inferring delays in partially observed gene regulatory networks*

2022-11-21, Seoul, Korea, **Population Approach Group Korea Annual Meeting**  
*Increasing viral transmission paradoxically reduces severe COVID-19 during endemic transition*

2022-10-19, Seoul, Korea, **Global KMS International Conference**  
*Systematic inference identifies a major source of heterogeneity in non-Markovian cell signaling dynamics*

2022-06-25, Yeosu, Korea, **KSMB Annual Meeting**  
*Increasing viral transmission paradoxically reduces severe COVID-19 during endemic transition*

2022-06-22, Online, **BRIC Webinar**  
*Systematic inference for cell signalling pathways identifies a key determinant of cell-to-cell variability*

2022-05-28, Daejeon, Korea, **KSIAM Spring Conference**  
*Systematic inference for cell signalling pathways identifies a key determinant of cell-to-cell variability*

2021-08-27, Online, Korea, **KSMB Annual Meeting**  
*Inference of stochastic dynamics in biochemical reaction networks*

2021-06-16, Online, **SMB Annual Meeting**  
*Inference of stochastic dynamics in biochemical reaction networks*

2021-05-27, Online, **SIAM Conference on Applications of Dynamical Systems**  
*Derivation of stationary distributions of stochastic chemical reaction networks via network translation*

2021-05-13, Online, **Seminar on the Mathematics of Reaction Networks**  
*Derivation of stationary distributions of stochastic chemical reaction networks via network translation [video]*

2020-10-24, Online, **KMS Annual Meeting, Minisymposium**  
*Derivation of stationary distributions of biochemical reaction networks via structure transformation*

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| ACADEMIC<br>SERVICE | <p><b>2021-06: Support Staff, SMB Annual Meeting</b><br/>Hosted Zoom sessions for the Asia time zone.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| PEER REVIEW         | <p>PLoS Computational Biology, Acta Biotheoretica, Stat, PCI Math Comp Biol, Frontiers in Public Health</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| OUTREACH            | <p><b>2026-05-26: Alumni Career Talk</b>, KAIST Department of Mathematical Sciences<br/> <i>Preparing for a Postdoc Abroad: A Three-Year Retrospective</i> [<a href="#"><i>slides</i></a>]<br/>         Slides are available but written in Korean.</p> <p><b>2023-06-20: KAIST Math Graduate Student Seminar</b>, KAIST Department of Mathematical Sciences<br/> <i>Preparation for Postdoc Positions</i></p> <p><b>2022-09-16: Science Slam-D Public Lecture</b>, Yongsan High School, Daejeon, Korea<br/> <i>Mathematics Meets Biology: Discovering the Optimal Sleep Pattern through Mathematical Modeling</i><br/>         Hosted by NST, IBS, Daejeon MBC (TV network), and Hello DD (Press). Delivered for high school students.</p> <p><b>2021-04-27: Mobile Science Classroom</b>, National Research Foundation of Korea<br/> <i>Mathematical Biology: Bridging Mathematics and the Life Sciences</i><br/>         Delivered for middle school students.</p> |